

IT-SIMPLERA INSTITUTE

Cyber Security Analyst Internship Program



WEEK 04 ASSIGNMENT REPORT

Cybersecurity Fundamentals & Security Concepts

Reconnaissance, OSINT & Attack Surface Assessment

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Assigned To: Cyber Security Analyst Interns

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Introduction

This report is submitted as part of the Week 04 assignment of the Cyber Security Analyst Internship Program at IT-SIMPLERA Institute. The assignment builds on the reconnaissance and Open Source Intelligence (OSINT) techniques covered in Week 3 and applies them to a real, authorized target.

The main goal of this task is to collect publicly available information about a target organization, map its attack surface, and record the findings in a clear and professional way — without performing any active exploitation or unauthorized testing.

Two authorized targets were used while completing the tasks in this report: shopify.com for the passive reconnaissance and Google Dork investigation (Tasks 1–3), and cloudflare.com for the subdomain enumeration, live host verification, and remaining tasks (Tasks 4–10). Both are large public organizations with bug bounty / responsible disclosure programs that allow this type of passive reconnaissance.

Task 1 – Passive Reconnaissance

Purpose: To collect publicly available information about the target (organization details, domain, hosting, SSL, and email security) without directly touching or scanning the live system.

Selected target: shopify.com

- **Organization overview**

The screenshot shows a search engine interface with a dark theme. At the top, it says 'About this page' with a share icon and a close button. Below that is the Shopify logo and the text 'Shopify Pakistan'. A blue button labeled 'Visit >' is to the right. The main text describes Shopify as 'The world's best commerce platform. ... Sell everywhere people shop. Online and in person. Across AI and on social. Locally and globally.' Below this are three buttons: 'Share', 'Save', and 'Remove result >'. A section titled 'About the source' is at the bottom, with a red box highlighting the text 'Shopify' and 'E-commerce company'. Below this, a paragraph describes Shopify Inc. as a Canadian multinational e-commerce company headquartered in Ottawa, Ontario, with over 5 million customers and \$292.3 billion in transactions in 2024, 57% of which was in the United States. A link to 'Wikipedia' is provided.

About this page

 Shopify

Shopify Pakistan [Visit >](#)

The world's best commerce platform. ... Sell everywhere people shop. Online and in person. Across AI and on social. Locally and globally.

[Share](#) [Save](#) [Remove result >](#)

About the source

Shopify
E-commerce company

Shopify Inc. is a Canadian multinational e-commerce company headquartered in Ottawa, Ontario that operates a platform for retail point-of-sale systems. The company has over 5 million customers and processed US\$292.3 billion in transactions in 2024, of which 57% was in the United States. [Wikipedia](#)

- **Main domain**
Shopify.com

- **Subdomains**

Change	Subdomain	IP	Cloudflare
UPDATED	changelog.shopify.com	23.227.39.20 34.128.177.21	
UPDATED	cms.everest.shopify.com	104.18.33.152 172.64.154.104	
UPDATED	collabs-m-staging.shopify.com	104.18.32.15 172.64.155.241	
UPDATED	collabs-m.shopify.com	104.18.32.15 172.64.155.241	
UPDATED	content.shopify.com	104.18.35.154 172.64.152.102	
UPDATED	everest.shopify.com	104.18.33.152 172.64.154.104	
UPDATED	inbox.shopify.com	34.128.177.21 23.227.39.20	
UPDATED	join.collabs.shopify.com	172.64.155.241 104.18.32.15	
UPDATED	mail.shopify.com	142.251.110.121 192.178.183.121	
UPDATED	mc-proxy.shopify.com	172.64.155.241 104.18.32.15	
UPDATED	status.shopify.com	18.239.69.104 18.239.69.85	

- **IP ownership(ASN) , Registrar , Hosting provider**

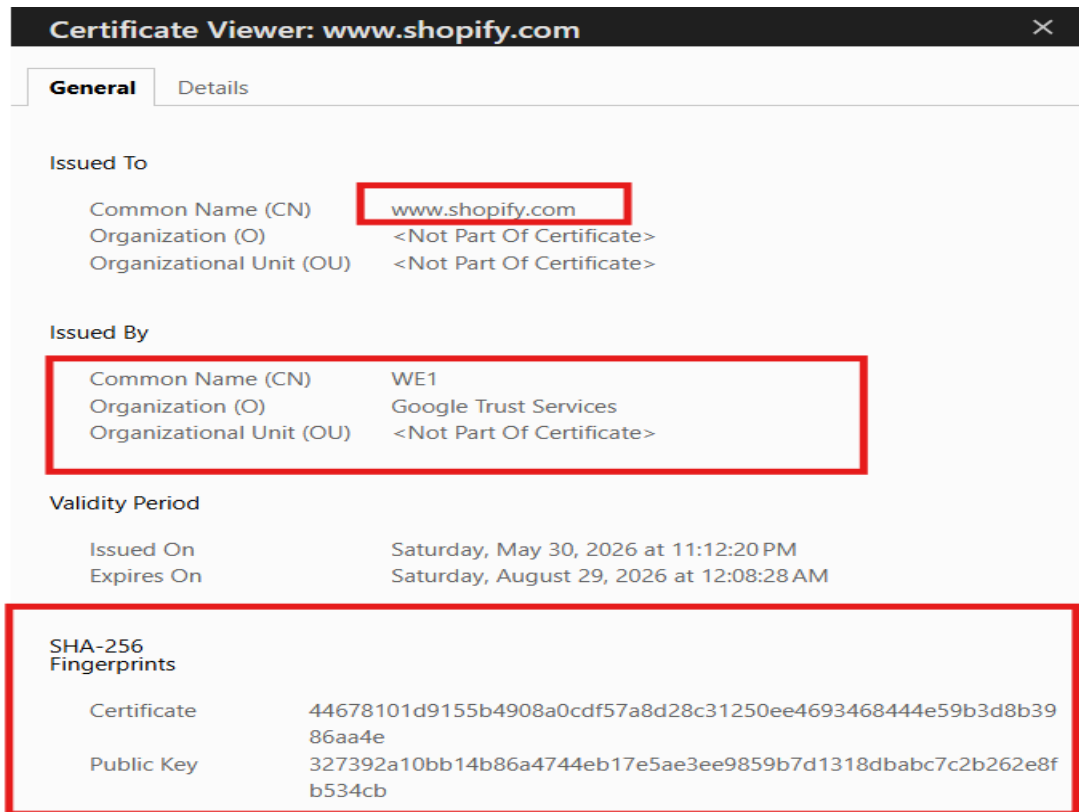
```
-----
[*] IP & ASN Information
-----
IP Address      : 23.227.38.33
ASN             : 13335
ASN Description : CLOUDFLARENET - Cloudflare, Inc., US
Hosting Provider: SHOPIFY-NET
```

- **Technologies identified**

```
-----
[*] HTTP Security Headers & Technologies
-----
Strict-Transport-Security: max-age=15552000; includeSubDomains; preload
Content-Security-Policy  : Not Found
X-Frame-Options          : Not Found
X-Content-Type-Options   : nosniff
Referrer-Policy          : Not Found

[Technologies / Fingerprinting (Best Effort)]
Server                   : cloudflare
X-Powered-By            : Unknown
```

- **SSL certificate information**



- **Email security records (SPF, DKIM, DMARC where applicable)**

```
[Email Security: SPF Record]
- Found: v=spf1 include:_spf.google.com include:mail.zendesk.com include:sendgrid.net ~all

[Email Security: DMARC Record]
- Found: "v=DMARC1; p=reject; pct=100; fo=1; rua=mailto:dmarc-aggregate@shopify.com;ruf=mailto:dmarc-reports@shopify.com"

[Email Security: DKIM Record (Best-Effort Selector Check)]
- [google] "v=DKIM1; k=rsa; p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCYb9CEvB9F144NTJtFQAlzmnP5ZJ7T8kdJ4QV58E/GBkwDQPCvajrwxk
QubkTE7MfRF2Rk5t0b0vy0j+nj/LiftFFdqi1UUCQ9Rie79JaLTYRRfhd4FeEmt6IoEf9bjio0qVz4Eao5MgXX1asdUhc9fQIDAQAB"
- [mail] "v=DKIM1; g=*; k=rsa; p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCYb9CEvB9F144NTJtFQAlzmnP5ZJ7T8kdJ4QV58E/GBkwDQPCvajrwxk
tWhKhKZbXLkH6GIdAbe2u8e3cSpfHZN5kvDqDL7w6Kj+aW0EQX6w0Prt6lWdgz9W8vdD7TCfXH3BrR0h05qX0YP3yVwYXRqpIQIDAQAB"
```

Task 2 – Advanced WHOIS & DNS Enumeration

Purpose: To use WHOIS and DNS lookup tools to study domain ownership, registration dates, name servers, and DNS records (A, AAAA, MX, NS, TXT, CNAME, SOA).

```
-----
[*] WHOIS Information
-----
Organization   : Shopify Inc.
Registrar      : MarkMonitor, Inc.
Creation Date   : 2005-03-11 05:18:03
Expiration Date : 2027-03-11 05:18:03
Name Servers    : GOLD.FOUNDATIONDNS.COM, GOLD.FOUNDATIONDNS.NET, GOLD.FOUNDATIONDNS.ORG, gold.foundationdns.org, gold.
ndationdns.com
```

[*] DNS Records

[A Records]

- 23.227.38.33

[AAAA Records]

- Not Found

[MX Records]

- 5 alt2.aspmx.l.google.com.
- 1 aspmx.l.google.com.
- 10 alt3.aspmx.l.google.com.
- 10 alt4.aspmx.l.google.com.
- 5 alt1.aspmx.l.google.com.

[NS Records]

- gold.foundationdns.org.
- gold.foundationdns.com.
- gold.foundationdns.net.

[TXT Records]

- "v=spf1 include:_spf.google.com include:mail.zendesk.com include:sendgrid.net ~all"
- "yahoo-verification-key=9t6Xys7YEajKycpYmHz742ZV/lm84njkfUfGuld0tZM="
- "zapier-domain-verification-challenge=506c545d-443f-4439-8d84-64648211b1f1"
- "00078847"
- "0lc931fl5ld2dpl15vx2flfkdwvrrzx"
- "MS=ms27001972"
- "_globalsign-domain-verification=1x11a1Wg3i08rScgK7aAMUJm_fdzKxH4whkWxr1bbg"
- "_globalsign-domain-verification=_PJNysq_1XyZleC1yx45rb_EUgbgkaJU36yk3CK0tk"
- "adobe-idp-site-verification=45576d365ff5492a15bc403c11382aad83403810324300fd62c5f196ce9e9063"
- "amazonses:CxA00EM1odef6TrFP0hdQh/2R7RoZYy8YLYHk2Frk="
- "apple-domain-verification=eMDCoIZdcJThX3yQ"
- "atlassian-domain-verification=aakXh8UjEwy75X6ck4l8jTIJDWHQ9CIunnUsE00mRgxM8IUzaJGLIt8zIINLALR"
- "autodesk-domain-verification=e9Nbi4FUDsS7clWU8iNj"
- "bitrise-verification=990e159e8448fcb6-hcIxnrbFeH9"
- "ca3-86f15a314f3342baac2abe7a8849163c"
- "ca3-fc9272b0aba34ba6991c0a62bc1998a0"
- "docuSign=de9614db-a0fa-4060-a1fa-2c444429ed9a"
- "drift-domain-verification=9b23e1f43b57171c988b4510e75a686bda241639096f7eaabe8ff3d72ac63476"
- "dtm-domain-verification=LhGwRr1DLWJoF6bVodh035n_BRj4Xu8PaEZ6bvvgd2Y"
- "facebook-domain-verification=u17rffysxyek688vqh4s02307suaza"
- "globalsign-domain-verification=Cpw4z0AIT5WSnINjnI4gafPgxsCJLEF4Ac_70Xm93I"
- "globalsign-domain-verification=lfM-pzwumuFWKV-wNEGq7a3KtsmRLJKsDxZzQaMjkw"
- "google-site-verification=0nU18bx0ue6doDAWZDnftfd66kTtGhWw00fdhflSd9es"

[Email Security: SPF Record]

- Found: v=spf1 include:_spf.google.com include:mail.zendesk.com include:sendgrid.net ~all

[Email Security: DMARC Record]

- Found: "v=DMARC1; p=reject; pct=100; fo=1; rua=mailto:dmarc-aggregate@shopify.com;ruf=mailto:dmarc-reports@shopify.com"

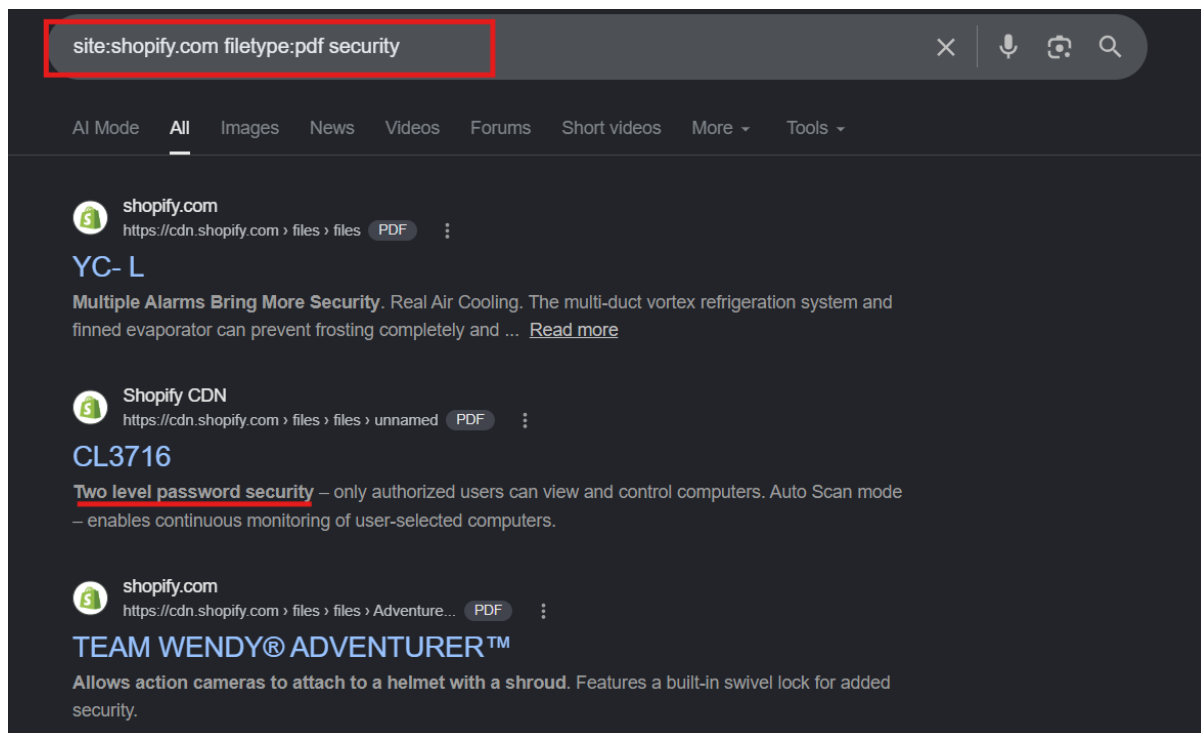
[Email Security: DKIM Record (Best-Effort Selector Check)]

- [google] "v=DKIM1; k=rsa; p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCYb9CEvB9F144NTJtFQAlzmnP5ZJ7T8kdJ4QV58E/GBkwDQPCvajrwxkQubkTE7MfRF2Rk5t0b0vy0j+nj/LiftFFdq1UUCQ9Rie79JaLtyRRfhd4FeEmt6IoEf9bjio0qVz4Eao5MgXKX1asdUhc9fQIDAQAB"
- [mail] "v=DKIM1; g=*; k=rsa; p=MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQCFVv1tmx0ilnHUpvpqpf4jlvCidUZL9A9NmMvzBifAtmn8Qxlfvyr tWhKhKZbXLH6GIudAbe2u8e3cSpfHZN5kvDqiDL7w6Kj+aW0EQX6w0Prt6lWdgz9W8vdD7TCfXH3BrR0h05qX0YP3yVwYXRqpIQIDAQAB"

Task 3 – Google Dork Investigation

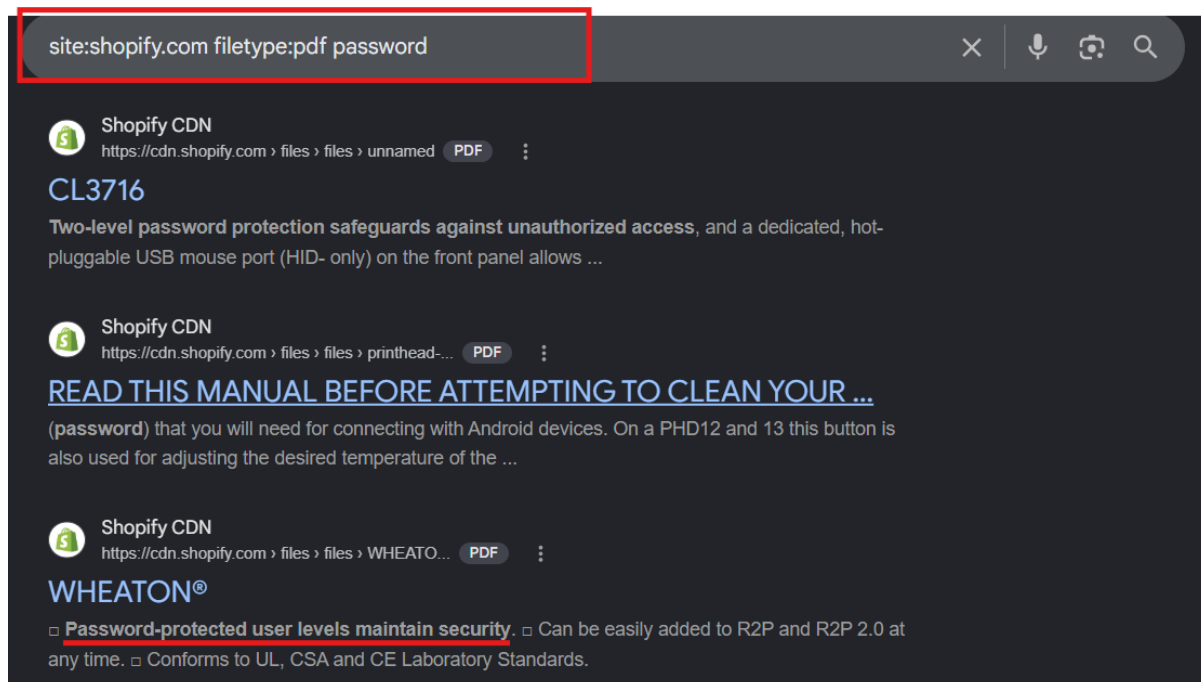
Purpose: To use advanced Google search operators (Google Dorks) to discover publicly indexed files, login pages, and documentation that may reveal useful or sensitive information about the target.

1. site:shopify.com filetype:pdf security



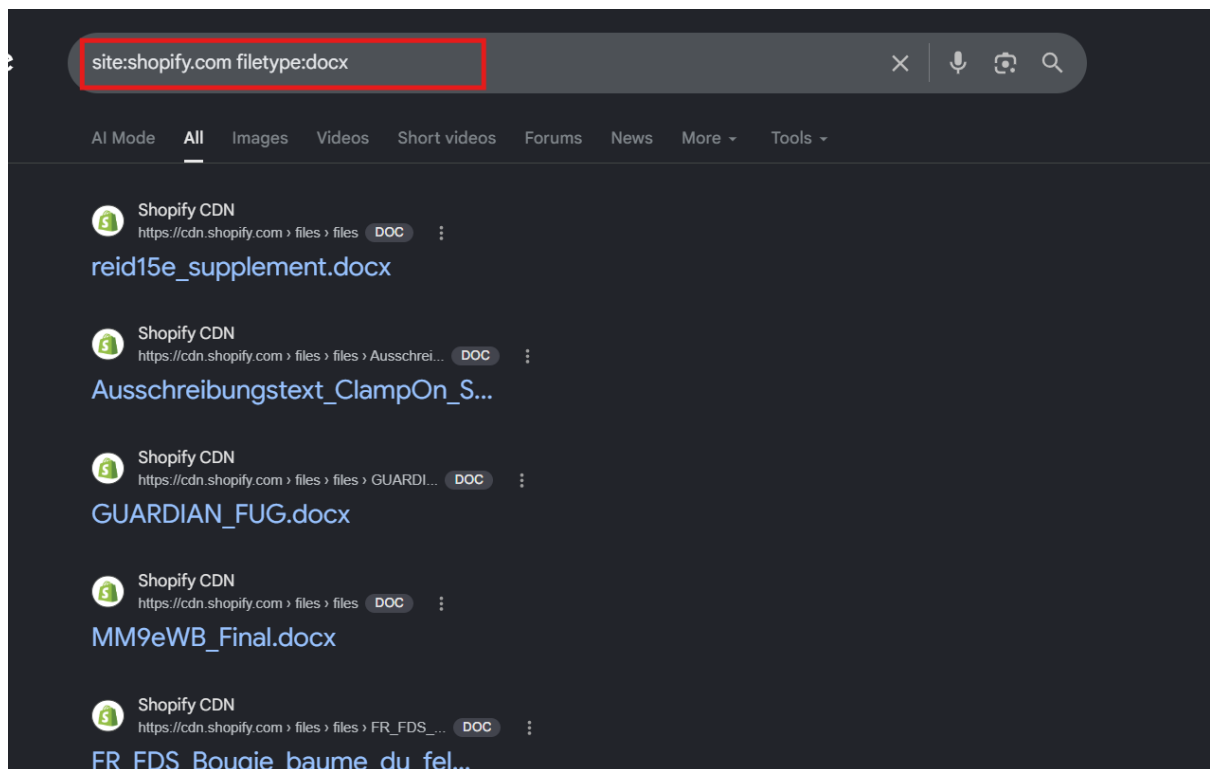
The screenshot shows a Google search interface with the query 'site:shopify.com filetype:pdf security' entered in the search bar. The search results are displayed on a dark background. The first result is from 'shopify.com' with the URL 'https://cdn.shopify.com › files › files' and a PDF icon. The title is 'YC- L' and the snippet mentions 'Multiple Alarms Bring More Security. Real Air Cooling. The multi-duct vortex refrigeration system and finned evaporator can prevent frosting completely and ...'. The second result is from 'Shopify CDN' with the URL 'https://cdn.shopify.com › files › files › unnamed' and a PDF icon. The title is 'CL3716' and the snippet mentions 'Two level password security – only authorized users can view and control computers. Auto Scan mode – enables continuous monitoring of user-selected computers.'. The third result is from 'shopify.com' with the URL 'https://cdn.shopify.com › files › files › Adventure...' and a PDF icon. The title is 'TEAM WENDY® ADVENTURER™' and the snippet mentions 'Allows action cameras to attach to a helmet with a shroud. Features a built-in swivel lock for added security.'

2. site:shopify.com filetype:pdf password

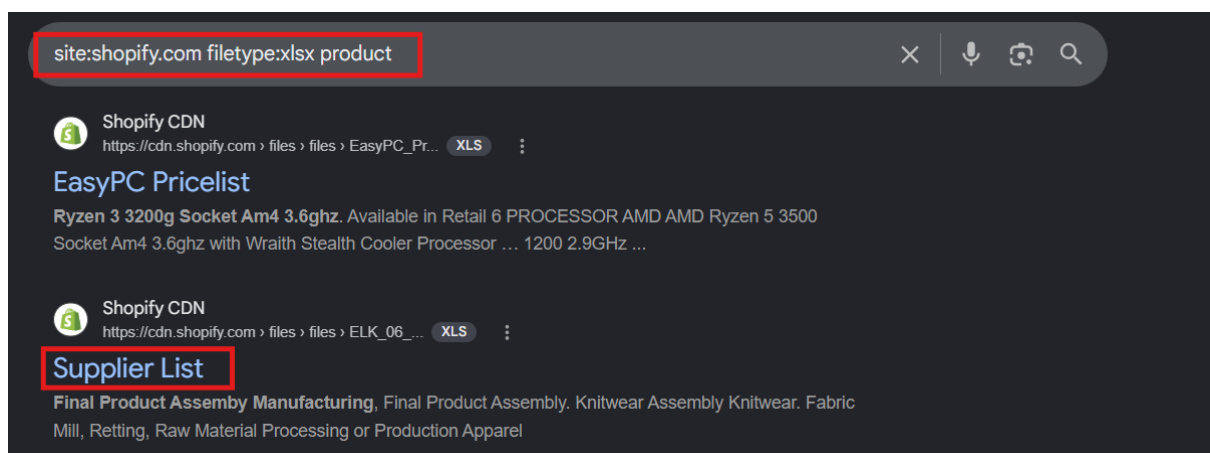


The screenshot shows a Google search interface with the query 'site:shopify.com filetype:pdf password' entered in the search bar. The search results are displayed on a dark background. The first result is from 'Shopify CDN' with the URL 'https://cdn.shopify.com › files › files › unnamed' and a PDF icon. The title is 'CL3716' and the snippet mentions 'Two-level password protection safeguards against unauthorized access, and a dedicated, hot-pluggable USB mouse port (HID- only) on the front panel allows ...'. The second result is from 'Shopify CDN' with the URL 'https://cdn.shopify.com › files › files › printhead...' and a PDF icon. The title is 'READ THIS MANUAL BEFORE ATTEMPTING TO CLEAN YOUR ...' and the snippet mentions '(password) that you will need for connecting with Android devices. On a PHD12 and 13 this button is also used for adjusting the desired temperature of the ...'. The third result is from 'Shopify CDN' with the URL 'https://cdn.shopify.com › files › files › WHEATO...' and a PDF icon. The title is 'WHEATON®' and the snippet mentions 'Password-protected user levels maintain security. Can be easily added to R2P and R2P 2.0 at any time. Conforms to UL, CSA and CE Laboratory Standards.'

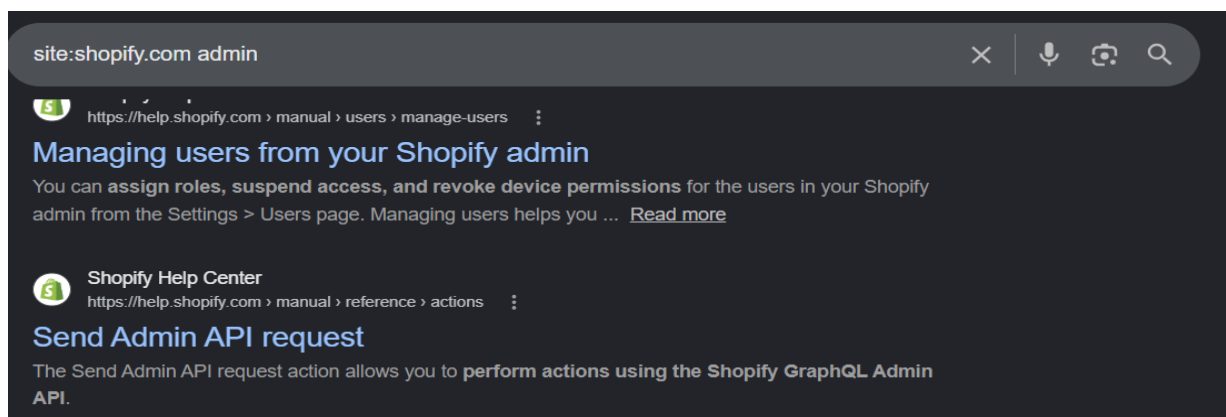
3. site:shopify.com filetype:docx



4. site:shopify.com filetype:xlsx



5. site:shopify.com admin



6. site:shopify.com API documentation

The screenshot shows a search engine interface with a search bar containing the query "site:shopify.com API documentation". Below the search bar, there are navigation tabs: "AI Mode", "All", "Images", "Videos", "News", "Forums", "Short videos", "More", and "Tools". The search results are displayed in a list format. The first result is from "Shopify" with the URL "https://www.shopify.com > blog > ecommerce-api" and the title "Ecommerce APIs: Types and Integration Guide (2026)". The second result is from "shopify.com" with the URL "https://developers.shopify.com" and the title "Shopify Dev Docs". The third result is from "Shopify Help Center" with the URL "https://help.shopify.com > getting-started > concepts > ad..." and the title "Shopify Flow and GraphQL Admin API".

Search query: **site:shopify.com API documentation**

Navigation: AI Mode | **All** | Images | Videos | News | Forums | Short videos | More | Tools

Results:

- Shopify**
https://www.shopify.com > blog > ecommerce-api
Ecommerce APIs: Types and Integration Guide (2026)
03-Oct-2025 — **API documentation** is fairly standardized and typically describes how developers should make a connection, send requests, and how data should be ... [Read more](#)
- shopify.com**
https://developers.shopify.com
Shopify Dev Docs
Learn how to build an app, theme, custom storefront, or marketplace. **Shopify's docs, dev tools and frameworks** make building easy and efficient. integrate into ...
- Shopify Help Center**
https://help.shopify.com > getting-started > concepts > ad...
Shopify Flow and GraphQL Admin API
API documentation to understand what is outputted by the API and used in Flow, and to ensure that your workflow outputs the data that you expect. Field ... [Read more](#)

7. site:shopify.com filetype:txt

The screenshot shows a search engine interface with a search bar containing the query "site:shopify.com filetype:txt". Below the search bar, there are navigation tabs: "AI Mode", "All", "Images", "Videos", "Short videos", "Forums", "News", "More", and "Tools". The search results are displayed in a list format. The first result is from "Shopify Hardware Store" with the URL "https://hardware.shopify.com > robots" and the title "robots.txt". The second result is from "Shopify" with the URL "https://www.shopify.com > robots" and the title "robots.txt". The third result is from "Shopify" with the URL "https://www.shopify.com > llms" and the title "llms.txt". The fourth result is from "Shopify" with the URL "https://www.shopify.com > ads" and the title "ads.txt". The fifth result is from "Shopify Help Center" with the URL "https://help.shopify.com > robots" and the title "robots.txt".

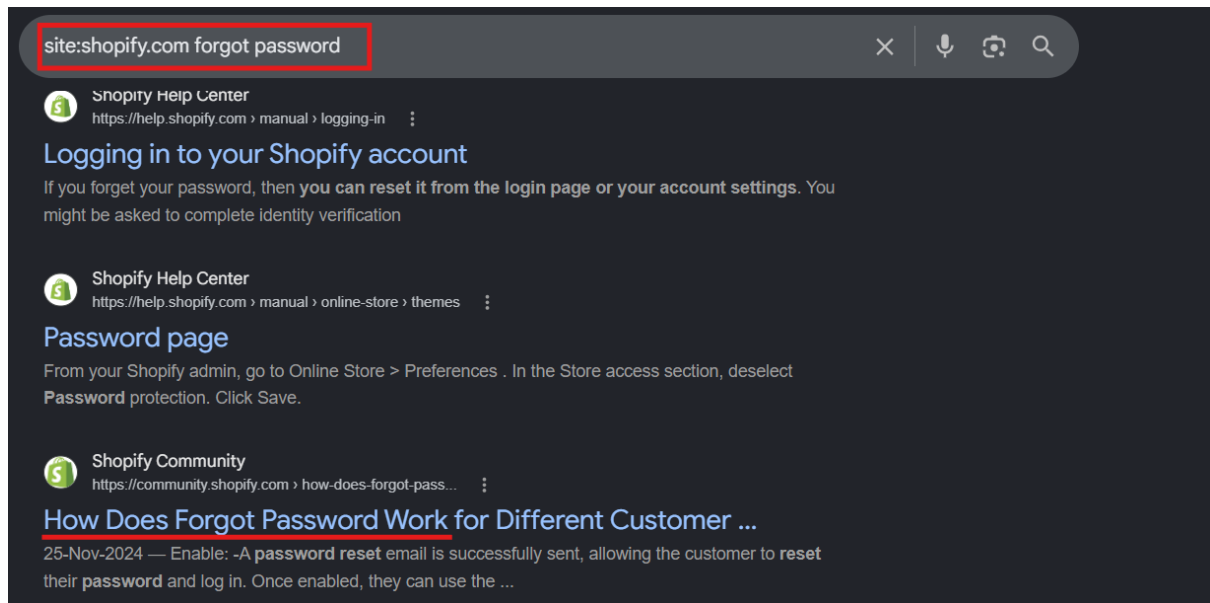
Search query: **site:shopify.com filetype:txt**

Navigation: AI Mode | **All** | Images | Videos | Short videos | Forums | News | More | Tools

Results:

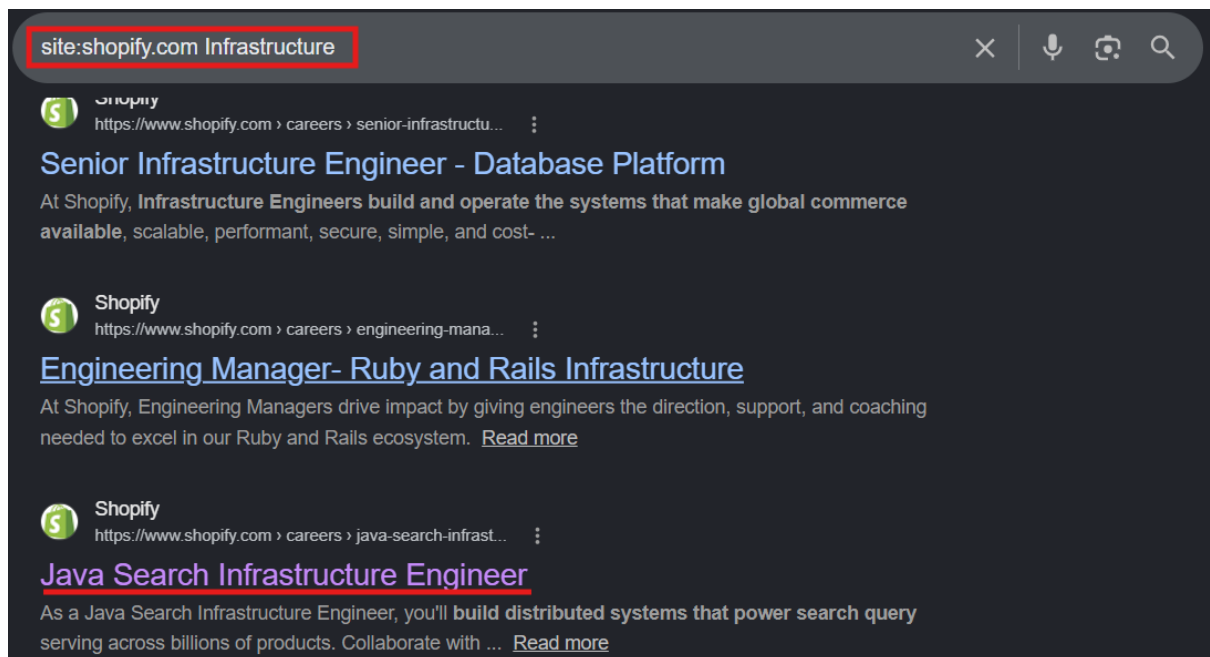
- Shopify Hardware Store**
https://hardware.shopify.com > robots
robots.txt
- Shopify**
https://www.shopify.com > robots
robots.txt
- Shopify**
https://www.shopify.com > llms
llms.txt
- Shopify**
https://www.shopify.com > ads
ads.txt
- Shopify Help Center**
https://help.shopify.com > robots · [Translate this page](#)
robots.txt

8. site:shopify.com forgot password



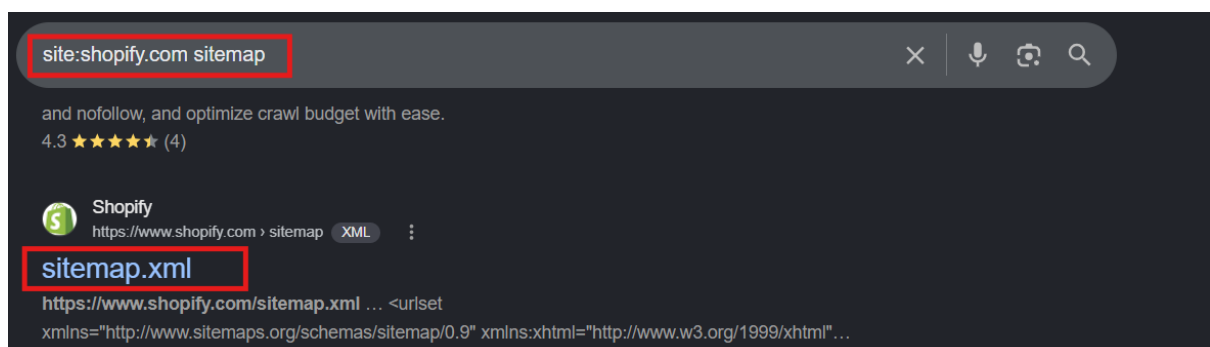
The screenshot shows search results for the query "site:shopify.com forgot password". The search bar at the top contains the query. The first result is from the "Shopify Help Center" with the URL "https://help.shopify.com › manual › logging-in". The title is "Logging in to your Shopify account" and the snippet says "If you forget your password, then you can reset it from the login page or your account settings. You might be asked to complete identity verification". The second result is also from the "Shopify Help Center" with the URL "https://help.shopify.com › manual › online-store › themes". The title is "Password page" and the snippet says "From your Shopify admin, go to Online Store > Preferences . In the Store access section, deselect Password protection. Click Save.". The third result is from the "Shopify Community" with the URL "https://community.shopify.com › how-does-forgot-pass...". The title is "How Does Forgot Password Work for Different Customer ..." and the snippet says "25-Nov-2024 — Enable: -A password reset email is successfully sent, allowing the customer to reset their password and log in. Once enabled, they can use the ...".

9. site:shopify.com Infrastructure



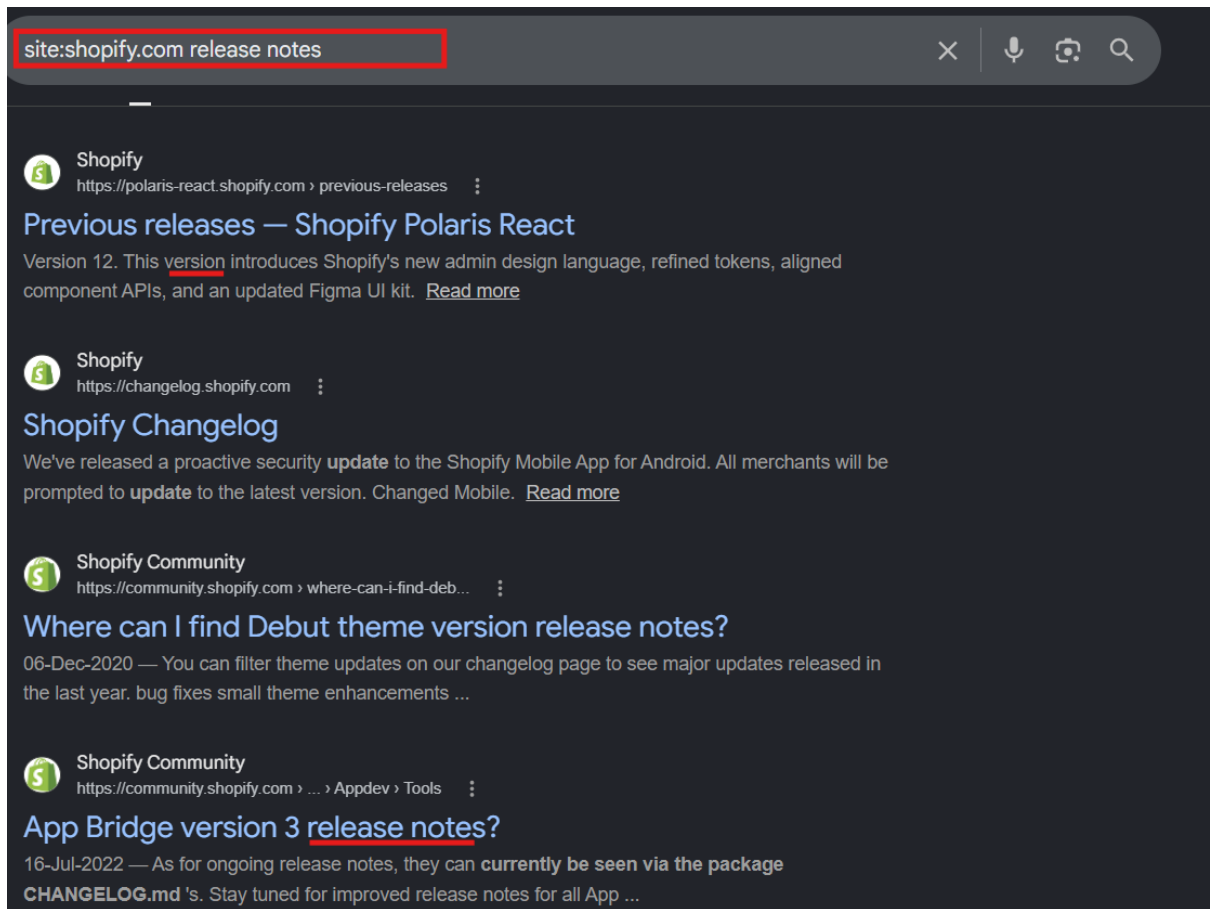
The screenshot shows search results for the query "site:shopify.com Infrastructure". The search bar at the top contains the query. The first result is from "Shopify" with the URL "https://www.shopify.com › careers › senior-infrastructu...". The title is "Senior Infrastructure Engineer - Database Platform" and the snippet says "At Shopify, Infrastructure Engineers build and operate the systems that make global commerce available, scalable, performant, secure, simple, and cost- ...". The second result is also from "Shopify" with the URL "https://www.shopify.com › careers › engineering-mana...". The title is "Engineering Manager- Ruby and Rails Infrastructure" and the snippet says "At Shopify, Engineering Managers drive impact by giving engineers the direction, support, and coaching needed to excel in our Ruby and Rails ecosystem. [Read more](#)". The third result is from "Shopify" with the URL "https://www.shopify.com › careers › java-search-infrast...". The title is "Java Search Infrastructure Engineer" and the snippet says "As a Java Search Infrastructure Engineer, you'll build distributed systems that power search query serving across billions of products. Collaborate with ... [Read more](#)".

10. Additional result for dork 11 (shows sitemap.xml publicly listed)



The screenshot shows search results for the query "site:shopify.com sitemap". The search bar at the top contains the query. The first result is a snippet "and nofollow, and optimize crawl budget with ease." with a rating of "4.3 ★★★★★ (4)". The second result is from "Shopify" with the URL "https://www.shopify.com › sitemap" and a file type indicator "XML". The title "sitemap.xml" is highlighted with a red box. The snippet shows the URL "https://www.shopify.com/sitemap.xml ... <urlset" and the XML namespace "xmlns='http://www.sitemaps.org/schemas/sitemap/0.9' xmlns:xhtml='http://www.w3.org/1999/xhtml' ...".

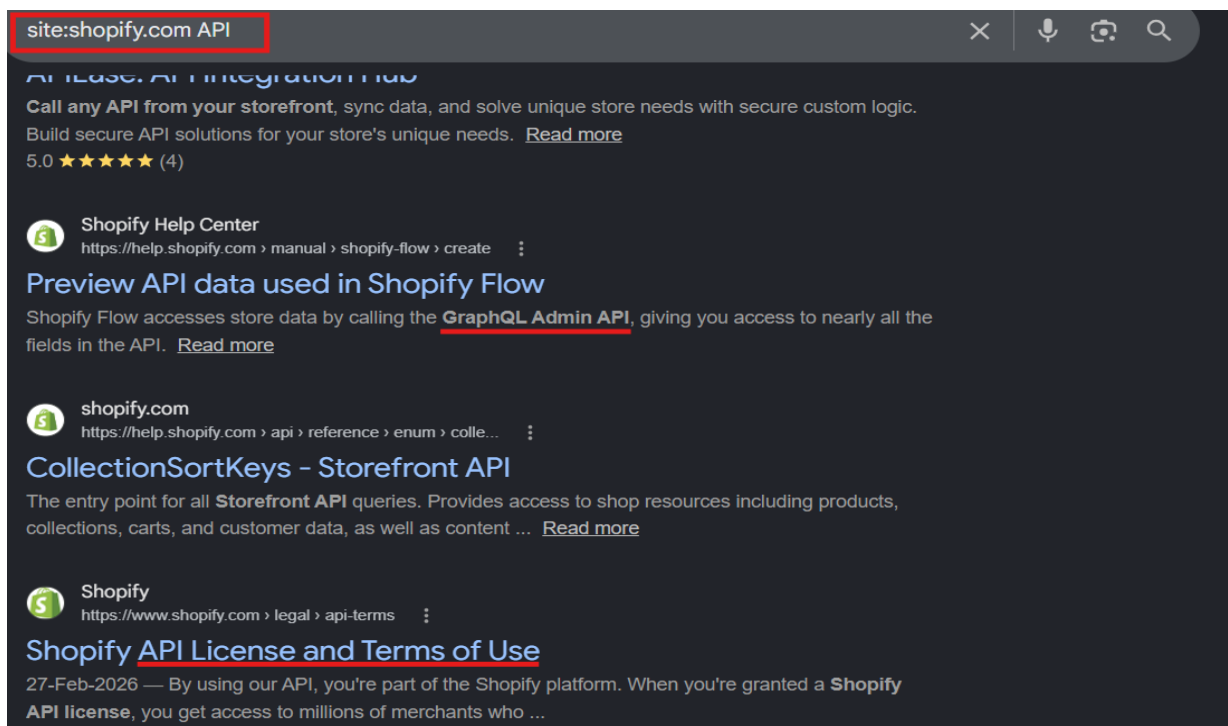
11. site:shopify.com release notes



The screenshot shows a search engine interface with the query "site:shopify.com release notes" in the search bar. Below the search bar, there are four search results, each with a Shopify logo icon, a title, a URL, and a brief description.

- Shopify**
https://polaris-react.shopify.com › previous-releases
Previous releases — Shopify Polaris React
Version 12. This version introduces Shopify's new admin design language, refined tokens, aligned component APIs, and an updated Figma UI kit. [Read more](#)
- Shopify**
https://changelog.shopify.com
Shopify Changelog
We've released a proactive security update to the Shopify Mobile App for Android. All merchants will be prompted to update to the latest version. Changed Mobile. [Read more](#)
- Shopify Community**
https://community.shopify.com › where-can-i-find-deb...
Where can I find Debut theme version release notes?
06-Dec-2020 — You can filter theme updates on our changelog page to see major updates released in the last year. bug fixes small theme enhancements ...
- Shopify Community**
https://community.shopify.com › ... › Appdev › Tools
App Bridge version 3 release notes?
16-Jul-2022 — As for ongoing release notes, they can currently be seen via the package **CHANGELOG.md** 's. Stay tuned for improved release notes for all App ...

12. site:shopify.com API



The screenshot shows a search engine interface with the query "site:shopify.com API" in the search bar. Below the search bar, there are four search results, each with a Shopify logo icon, a title, a URL, and a brief description.

- Shopify**
https://polaris-react.shopify.com › previous-releases
Previous releases — Shopify Polaris React
Version 12. This version introduces Shopify's new admin design language, refined tokens, aligned component APIs, and an updated Figma UI kit. [Read more](#)
- Shopify**
https://changelog.shopify.com
Shopify Changelog
We've released a proactive security update to the Shopify Mobile App for Android. All merchants will be prompted to update to the latest version. Changed Mobile. [Read more](#)
- Shopify Community**
https://community.shopify.com › where-can-i-find-deb...
Where can I find Debut theme version release notes?
06-Dec-2020 — You can filter theme updates on our changelog page to see major updates released in the last year. bug fixes small theme enhancements ...
- Shopify Community**
https://community.shopify.com › ... › Appdev › Tools
App Bridge version 3 release notes?
16-Jul-2022 — As for ongoing release notes, they can currently be seen via the package **CHANGELOG.md** 's. Stay tuned for improved release notes for all App ...

Google Dork Summary Table

#	Google Dork Query	Purpose	Result Summary
1	site:shopify.com filetype:pdf security	Look for PDF files that mention security topics.	Product manuals mentioning security features were found; no sensitive internal PDFs.
2	site:shopify.com filetype:pdf password	Look for PDF files that mention passwords.	Public product manuals referencing password protection were found; nothing sensitive.
3	site:shopify.com filetype:docx	Search for exposed Word documents.	See screenshot above for indexed results.
4	site:shopify.com filetype:xlsx	Search for exposed Excel spreadsheets.	See screenshot above for indexed results.
5	site:shopify.com API documentation	Locate public API documentation pages.	Shopify developer/API documentation pages were indexed and publicly accessible.
6	site:shopify.com filetype:txt	Search for exposed plain text files.	See screenshot above for indexed results.
7	site:shopify.com forgot password	Find the account/password recovery flow.	Shopify Help Center pages explaining the password-reset process were found.
8	site:shopify.com admin	Look for admin-related pages or portals.	Help Center pages about managing admin users were found; no exposed live admin panel.
9	site:shopify.com Infrastructure	Use career/job pages to learn about the tech stack.	Job postings for infrastructure engineers revealed some technology and team details.
10	site:shopify.com release notes	Find changelog / version history pages.	Public changelog and release-notes pages were

			found, showing update history.
11	site:shopify.com API	Locate additional pages related to the API.	Further API-related and sitemap pages were found, including a public sitemap.xml.

Task 4 – Subdomain Enumeration:

Purpose: To discover subdomains belonging to the target using multiple reconnaissance tools, then merge, clean, and categorize the results.

Target:cloudflare.com

Subdomain	Category	Purpose	Security Relevance
api. cloudflare.com	API	Backend APIs	Authentication, authorization, API security
blog.api. cloudflare.com	API / Blog	API documentation or blog	Public documentation, API information
assets. cloudflare.com	Assets/CDN	Static files	Asset delivery, caching
cf-assets. cloudflare.com	CDN	Cloudflare assets	Static content infrastructure
cf-obs. cloudflare.com	Observability	Monitoring/Telemetry	Logging and monitoring infrastructure
events. cloudflare.com	Events	Company events	Public-facing web application
fieldmarketing. cloudflare.com	Marketing	Campaign pages	Public web application
pages. cloudflare.com	Pages Platform	Cloudflare Pages	Static hosting platform
timeline. cloudflare.com	Timeline	Status/History	Operational information
willow. cloudflare.com	Internal project (unknown)	Unknown	Requires verification before categorizing

Task 5 – Live Host Verification

Purpose: To confirm which discovered subdomains are actually online, and to record their HTTP status, page title, server, and technology details.

Subdomain	HT TP Stat us	HTT PS	Page Title	Web Server	Technolo gy Fingerpr int	Notes
cloudflare.com	301	Yes	301 Moved Permane ntly	cloudfl are	Cloudflar e, Cloudflar e Bot Manage ment, HSTS, HTTP/3	Redirects to the primary website.
playground.ai.cloudflare.com	200	Yes	AI Playgrou nd	cloudfl are	Cloudflar e	Public AI playgroun d is accessible.
abuse.cloudflare.com	200	Yes	Abuse form Cloudflar e The web performa nce & security company	cloudfl are	Cloudflar e, OneTrust	Public abuse reporting portal.
multi- modal.ai.cloudflare.com	200	Yes	Cloudflar e Multi- modal Playgrou nd	cloudfl are	Cloudflar e	Public AI application .
agents.cloudflare.com	200	Yes	Cloudflar e Agents	cloudfl are	Astro 6.1.5, Cloudflar e	Public web application built with Astro.

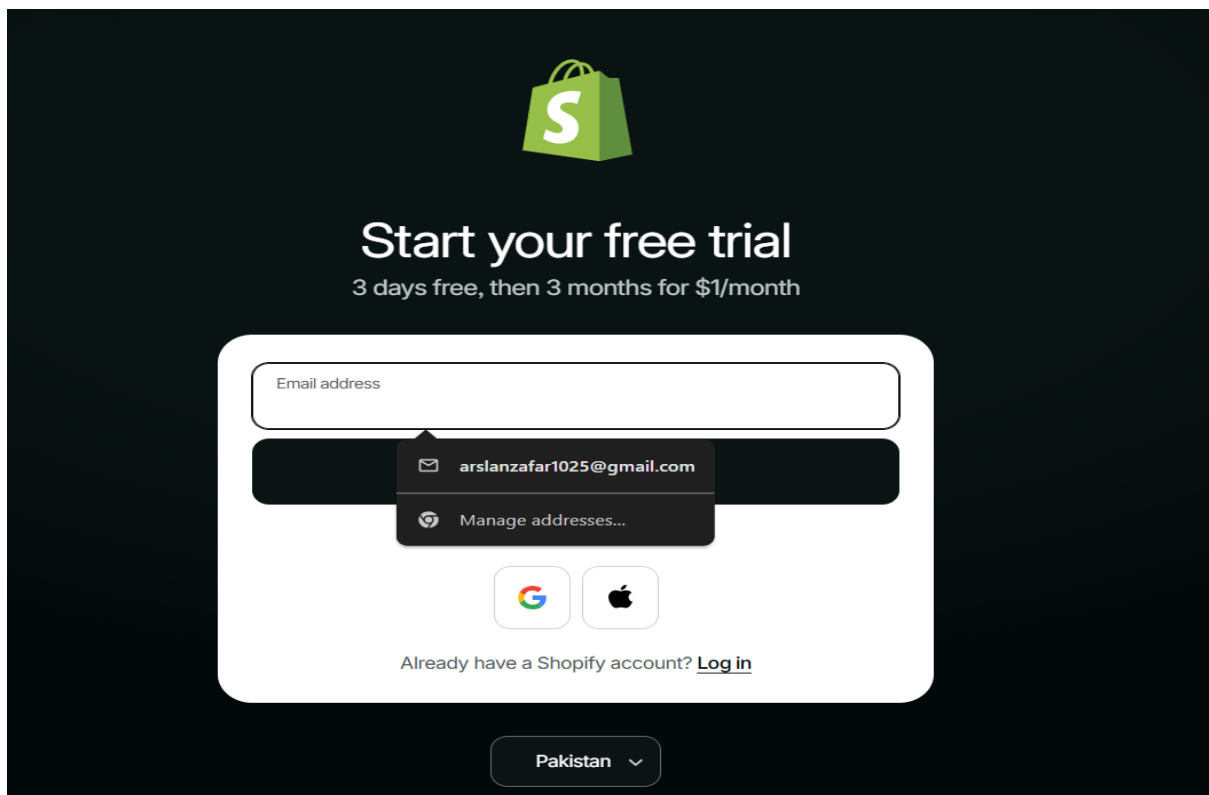
developers.cloudflare.com	200	Yes	Cloudflare Developer Docs Cloudflare Docs	cloudflare	Astro 7.1.3, Cloudflare, Cloudflare Bot Management, Cloudflare Browser Insights, HSTS, HTTP/3, OneTrust	Public developer documentation.
search.ai.cloudflare.com	200	Yes	AI Search Snippets – Embed AI Search on your website	cloudflare	Cloudflare, Cloudflare Browser Insights, cdnjs	Public AI search platform.
access.cloudflare.com	301	Yes	Redirect	cloudflare	Cloudflare	Redirects to another Cloudflare service.
ai.cloudflare.com	301	Yes	301 Moved Permanently	cloudflare	Cloudflare	Redirect to another AI-related endpoint.
aig.cloudflare.com	302	Yes	302 Found	cloudflare	Cloudflare, Cloudflare Bot Management	Temporary redirect.
gateway.ai.cloudflare.com	302	Yes	Redirect	cloudflare	Cloudflare, Cloudflare Bot Management	Redirect to gateway service.
websearch.ai.cloudflare.com	403	Yes	Forbidden	cloudflare	Cloudflare	Server is reachable but access is restricted.

panthera.run.ai.cloudflare.com	401	Yes	Unauthorized	cloudflare	Cloudflare	Authentication required.
stripe-app.provisioning.agents.cloudflare.com	401	Yes	Unauthorized	cloudflare	Cloudflare	Protected endpoint requiring authentication.
akd-auditor.cloudflare.com	404	Yes	Not Found	cloudflare	Cloudflare	Host is live, but requested resource is unavailable.

Task 6 – Web Screenshot Collection

Purpose: To visually capture and document the live web pages found during the assessment, including login pages, dashboards, and API portals.

Login Page:



Services:



Why Shopify ▾Products ▴PricingEnterprise

Spring '26 Edition

BUILD YOUR WEBSITE

Website Builder

Themes

Domains

Customer Accounts

Sidekick

RUN YOUR BUSINESS

Orders & Inventory

Shipping

Workflow Automation

Mobile

SELL ANYWHERE

Online

AI Chats

Point of Sale

Shop App

Social & Marketplaces

Global

B2B

Across Markets

MARKETING & ANALYTICS

Advertising & Campaigns

Email & Customer Chat

Discounts

Analytics

GET PAID

Checkout

Payments

CUSTOMIZE & EXTEND SHOPIFY

Commerce for Agents
Build with our agent tools

Shopify App Store
Largest commerce ecosystem

API documentation:

TABLE OF CONTENTS

What we value

1. Definitions

2. Using the Shopify API

3. Changes to the Shopify API

4. Proprietary Rights

5. Shopify Trademarks

6. Merchant Agreement, Privacy and Data Security

7. Security

What we value

By using our API, you're part of the Shopify platform. These are the values that we feel keep our platform strong.

These values aren't legally binding. To get the complete picture of our legal requirements, read the [Shopify API License and Terms of Use](#)



Respect our merchants

When you're granted a Shopify API license, you get access to millions of merchants who might be looking for an app just like yours. You'll still own your app, but Shopify owns the platform. As owners of the platform, we put reasonable limits on the use of the merchant data it allows you to access.

Only request the merchant data you need to provide your service, nothing more. Get merchants' permission to do so, and follow our rules for deleting personal customer data. If your public app captures customer data via a Shopify online store or POS,

Courses:

Looking for a place to start?

Whether you're new to Shopify or looking to grow your skills, we've got you covered.

View all

COURSE



Introduction to Shopify

Get to know Shopify: our mission, core features of the platform, and how we help businesses of all sizes start, run, and grow.

FREE28 MIN

COURSE



Navigating the Shopify Admin

Tour the Shopify admin with interactive demos and articles covering key features and tools.

FREE28 MIN

COURSE



Developing on Shopify

Get started on your journey developing on Shopify.

FREE68 MIN

Contact Form:



Get in touch with an expert.

- ✓ Get detailed pricing
- ✓ See how much you could save with Shopify
- ✓ Learn how Shopify adapts to your systems
- ✓ Get help migrating

Country
Select one

Annual online B2B revenue
Select one

Are you an agency contacting us on behalf of a brand?

☐ Yes ☒ No

Submit

Videos available at website:

TRACK

Shopify Developer

A Shopify Developer delivers Shopify Plus projects, from initial setup to merchant handover.

VIDEO

What is Shopify?

Learn how Shopify's unified commerce platform grows with every business. From first-time entrepreneurs to global...

FREE 2 MIN

VIDEO

Optimizing your Shopify Theme with PageSpeed

Discover why fast Shopify stores matter for both conversions and SEO. This video covers how to audit your Shopify store...

FREE 17 MIN

VIDEO

Building Accessible Themes

Create inclusive Shopify experiences through accessibility best practices: semantic HTML, keyboard nav., and...

FREE 15 MIN

Apps that they are selling

Get the best apps for your business with Built for Shopify

See why these shine >

Canva Connect
4.8 ★ (385) • Free
Access your product images and files directly inside Canva
+ Built for Shopify

UpPromote Affiliate Marketing
4.9 ★ (3,550) • Free to install
Drive referral sales loops with influencer & affiliate program
+ Built for Shopify

Instafeed - Instagram Feed
4.9 ★ (1,896) • Free plan available
Display Instagram feeds, Reels & Insta UGC as shoppable videos
+ Built for Shopify

Faire: Buy Wholesale
4.9 ★ (1,138) • Free
Buy wholesale from independent brands worldwide
+ Built for Shopify

Tiny SEO Speed Image Optimizer
5.0 ★ (2,226) • Free to install
Boost SEO traffic, AEO/AI search, page speed & minify images! ↑
+ Built for Shopify

Judge.me Product Reviews App
5.0 ★ (42,155) • Free plan available
Collect unlimited product reviews, star ratings, testimonials
+ Built for Shopify

Dev Community:

 dev community

Sign UpLog In🔍☰

Shopify community for partners and developers



categories >tags >

LatestCategoriesTop

Topic	Replies	Views	Activity
Next Generation Events now available in Developer Preview If you've used webhooks, you know the problem. You subscribe to products/update and now yo... Webhooks and Events	 39	1.1k	9m
Shopify billing for connector app? Hello together! We're currently developing a connector app, that's sole purpose is to connect Sh... Built for Shopify	 13	502	15m

Error Page:

404

Page not found

Time is money, so we won't waste yours. This page doesn't exist. Check out the [Help Center](#) to find what you're looking for.

Start

Pricing

Website Builder

Blog

→

→

→

→

Task 7 – Technology Fingerprinting

Purpose: To identify the web server, framework, CMS, JavaScript libraries, CDN, and WAF used by each live asset.

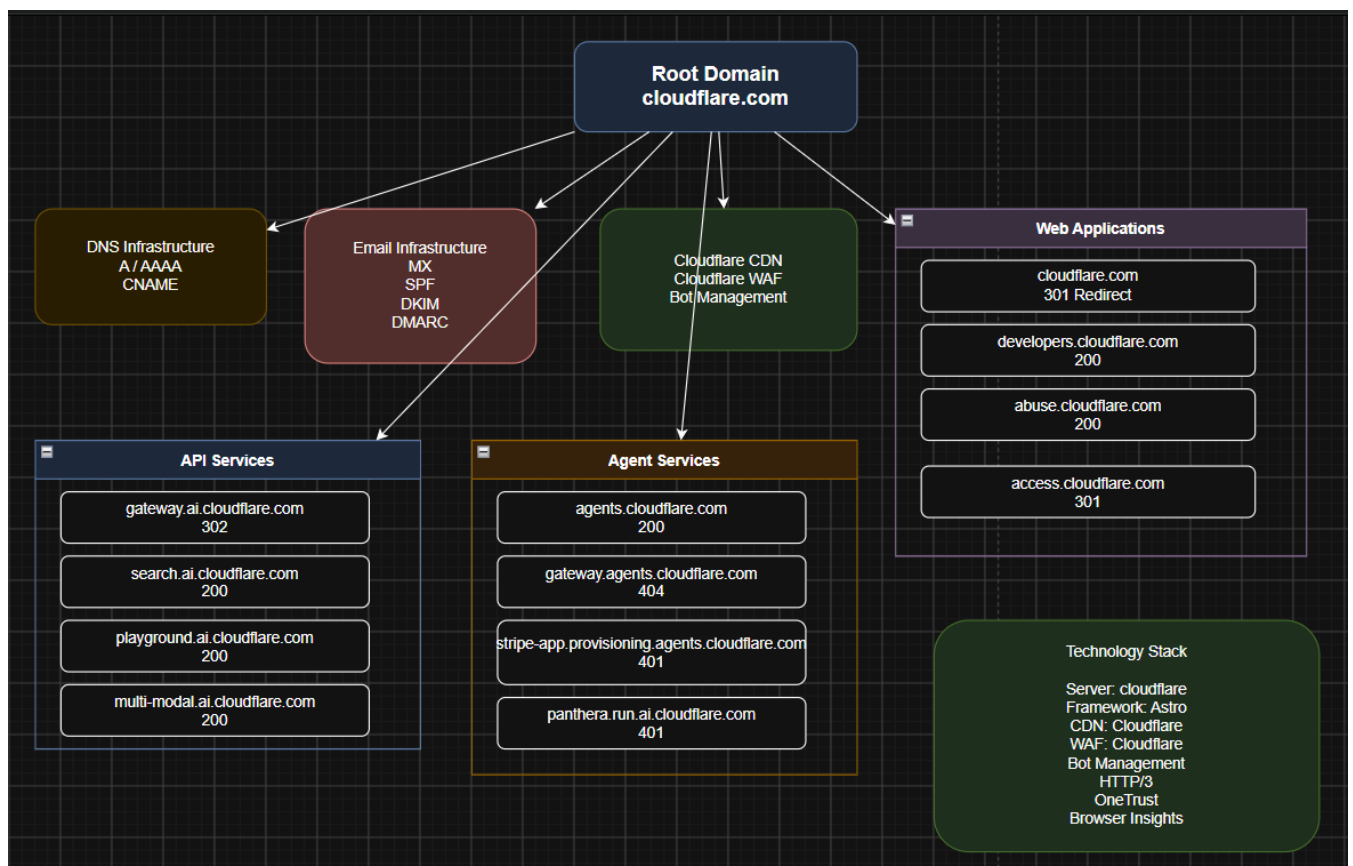
Live Asset	Web Server	Framework	CMS	JavaScript Libraries / Tools	CDN	WAF
cloudflare.com	cloudflare	Not Identified	None Detected	Cloudflare Browser Insights	Cloudflare	Cloudflare
playground.ai.cloudflare.com	cloudflare	Not Identified	None Detected	Not Identified	Cloudflare	Cloudflare
abuse.cloudflare.com	cloudflare	Not Identified	None Detected	OneTrust	Cloudflare	Cloudflare
multi-modal.ai.cloudflare.com	cloudflare	Not Identified	None Detected	Not Identified	Cloudflare	Cloudflare
agents.cloudflare.com	cloudflare	Astro 6.1.5	None Detected	Not Identified	Cloudflare	Cloudflare
developers.cloudflare.com	cloudflare	Astro 7.1.3	None Detected	Cloudflare Browser Insights, OneTrust	Cloudflare	Cloudflare
search.ai.cloudflare.com	cloudflare	Not Identified	None Detected	Cloudflare Browser Insights, cdnjs	Cloudflare	Cloudflare
access.cloudflare.com	cloudflare	Not Identified	None Detected	Not Identified	Cloudflare	Cloudflare
gateway.ai.cloudflare.com	cloudflare	Not Identified	None Detected	Not Identified	Cloudflare	Cloudflare

Technology Summary:

Technology	Observation
Web Server	Cloudflare web server was identified on all tested assets.
Programming Language	Could not be identified through passive fingerprinting.
Framework	Astro framework detected on <code>agents.cloudflare.com</code> and <code>developers.cloudflare.com</code> .
CMS	No Content Management System was detected.
JavaScript Libraries	Cloudflare Browser Insights, OneTrust and cdnjs were identified on some assets.
CDN	Cloudflare CDN was consistently detected across all assets.
WAF	Cloudflare Web Application Firewall (WAF) appears to protect the tested assets.

Task 8 – Attack Surface Mapping

Purpose: To combine all reconnaissance findings into one diagram that visually maps the target's domains, subdomains, and infrastructure.



Task 9 – Risk Identification:

Purpose: To review all reconnaissance findings and list the potential security risks they could lead to, along with a recommended fix for each one.

#	Possible Risk	Why it is a Risk	Recommendation
1	Public API Endpoints	Public APIs can be tested by attackers. If security is weak, they may access data or misuse the API.	Use authentication and check APIs regularly.
2	Development or Staging Servers	Test servers may contain unfinished code or test data.	Keep development and staging servers private or protect them with login.
3	Public Documentation	Documentation may give information about APIs or website structure.	Do not publish sensitive information in public documents.
4	Redirects (301/302)	Many redirects may reveal how the website is organized.	Check that redirects only go to trusted pages.
5	Login or Protected Pages (401)	Attackers can see that a login page exists and may try password attacks.	Use strong passwords and Multi-Factor Authentication (MFA).
6	Not Found Pages (404)	Many 404 pages may show old or unused subdomains.	Remove unused subdomains and old DNS records.
7	Technology Information	Tools can detect technologies like Astro or Cloudflare. Attackers may use this information during reconnaissance.	Keep software updated and avoid showing unnecessary information.
8	AI Services	Public AI services should be protected from misuse.	Add authentication and rate limiting where needed.
9	Email Security	Missing or weak SPF, DKIM, or DMARC records can increase the risk of email spoofing.	Configure SPF, DKIM, and DMARC correctly.
10	Large Attack Surface	Many public subdomains mean there are more systems to manage and protect.	Regularly review and remove unused subdomains.

Task 10 – Reconnaissance Automation:

Purpose: To automate part of the reconnaissance workflow with a Python script, saving time and making the process repeatable.

```
(kali㉿kali)-[~]
└─$ python3 tool.py

      ____          _              )
     /  __ \    ___/  |__  _____)_____
    /  /  \  /   /_  |__/_/_____/_____)_____/
   /___/\_\/___/___|_|_____/_____/_____/_____/
  Passive Reconnaissance Tool (Educational Use Only)

Enter Domain (e.g., shopify.com): cloudflare.com

[*] Starting Passive Reconnaissance on: cloudflare.com


-----
[*] WHOIS Information
-----
Organization       : DATA REDACTED
Registrar          : Cloudflare, Inc.
Creation Date      : 2009-02-17 22:07:54
Expiration Date    : 2033-02-17 22:07:54
Name Servers       : NS3.CLOUDFLARE.COM, NS4.CLOUDFLARE.COM, NS5.CLOUDFLARE.COM, NS6.CLOUDFLARE.COM, NS7.CLOUDFLARE.COM, ns3.cloudflare.co
m, ns4.cloudflare.com, ns5.cloudflare.com, ns6.cloudflare.com, ns7.cloudflare.com
```

```
[*] IP & ASN Information
-----
IP Address       : 104.16.132.229
ASN              : 13335
ASN Description  : CLOUDFLARENET - Cloudflare, Inc., US
Hosting Provider: CLOUDFLARENET
-----

[*] DNS Records
-----

[A Records]
- 104.16.133.229
- 104.16.132.229

[AAAA Records]
- 2606:4700::6810:84e5
- 2606:4700::6810:85e5

[MX Records]
- 5 mxb-canary.global.inbound.cf-emailsecurity.net.
- 10 mxa.global.inbound.cf-emailsecurity.net.
- 10 mxb.global.inbound.cf-emailsecurity.net.
- 5 mxa-canary.global.inbound.cf-emailsecurity.net.

[NS Records]
- ns5.cloudflare.com.
- ns6.cloudflare.com.
- ns7.cloudflare.com.
- ns3.cloudflare.com.
- ns4.cloudflare.com.

[TXT Records]
- "google-site-verification=C7thfNeXVahkVhniiqTI1iSVnElKR_kBBtnEHkeGDlo"
- "google-site-verification=ZdlQZLBbAPkxeFTCM1rpiB_ibtGff_JF5KllNKwDR9I"
- "liveramp-site-verification=EhH1MqgwbndTWl1AN64h0TKz7hc1s80vUpchLbgpfY0"
```

Code:

```
#!/usr/bin/env python3
"""
Passive Reconnaissance Tool (Educational Use Only)
Collects publicly available information about a given domain without active scanning.
Requires: pip install requests python-whois dnspython ipwhois
"""

import time
import socket
import ssl
import requests
import whois
import dns.resolver
from ipwhois import IPWhois
from urllib.parse import urlparse

# Define a summary dictionary to track success/failure of each module
summary = {
    "WHOIS": False,
    "DNS": False,
    "IP_ASN": False,
    "SSL": False,
    "HTTP_Security": False,
    "Web_Files": False
}

def print_section_header(title):
    print(f"\n{'-'*40}")
    print(f"[*] {title}")
    print(f"{'-'*40}")

def format_date(date_obj):
    if not date_obj:
        return "Unavailable"
    if isinstance(date_obj, list):
        return str(date_obj[0])
    return str(date_obj)

def get_whois_info(domain):
    print_section_header("WHOIS Information")
```



```

try:
    w = whois.whois(domain)
    org = w.org if w.org else w.name

    print(f"Organization : {org if org else 'Not Found'}")
    print(f"Registrar : {w.registrar if w.registrar else 'Not Found'}")
    print(f"Creation Date : {format_date(w.creation_date)}")
    print(f"Expiration Date: {format_date(w.expiration_date)}")

    ns = w.name_servers
    if isinstance(ns, list):
        print(f"Name Servers : {', '.join(ns)}")
    elif ns:
        print(f"Name Servers : {ns}")
    else:
        print("Name Servers : Not Found")

    summary["WHOIS"] = True
except Exception as e:
    print(f"[!] Could not retrieve WHOIS information: {e}")

def get_dns_info(domain):
    print_section_header("DNS Records")
    records_to_check = ['A', 'AAAA', 'MX', 'NS', 'TXT']
    success = False

    for record in records_to_check:
        try:
            answers = dns.resolver.resolve(domain, record)
            print(f"\n[{record} Records]")
            for rdata in answers:
                print(f" - {rdata.to_text()}")
            success = True
        except dns.resolver.NoAnswer:
            print(f"\n[{record} Records]\n - Not Found")
        except dns.resolver.NXDOMAIN:
            print(f"\n[{record} Records]\n - Domain does not exist.")
            break
        except Exception:
            print(f"\n[{record} Records]\n - Unavailable")

# Specifically check DMARC
try:
    dmarc_domain = f"_dmarc.{domain}"
    answers = dns.resolver.resolve(dmarc_domain, 'TXT')
    print("\n[DMARC Record]")
    for rdata in answers:
        print(f" - {rdata.to_text()}")

```

```

except Exception:
    print("\n[DMARC Record]\n - Not Found")

if success:
    summary["DNS"] = True

def get_ip_asn_info(domain):
    print_section_header("IP & ASN Information")
    try:
        ip = socket.gethostbyname(domain)
        print(f"IP Address    : {ip}")

        try:
            obj = IPWhois(ip)
            results = obj.lookup_rdap(depth=1)

            asn = results.get('asn', 'Not Found')
            asn_desc = results.get('asn_description', 'Not Found')
            provider = results.get('network', {}).get('name', 'Not Found')

            print(f"ASN          : {asn}")
            print(f"ASN Description : {asn_desc}")
            print(f"Hosting Provider: {provider}")

            summary["IP_ASN"] = True
        except Exception as e:
            print(f"[!] Could not retrieve ASN/Hosting data: {e}")
            summary["IP_ASN"] = True # IP was found at least
    except socket.gaierror:
        print("[!] Could not resolve domain to an IP address.")

def get_ssl_info(domain):
    print_section_header("SSL Certificate")
    port = 443
    context = ssl.create_default_context()
    context.check_hostname = False
    context.verify_mode = ssl.CERT_NONE # We just want to read it, not strictly validate chain here

    try:
        with socket.create_connection((domain, port), timeout=5) as sock:
            with context.wrap_socket(sock, server_hostname=domain) as ssock:
                cert = ssock.getpeercert(binary_form=False)

                # ssl dict parsing is a bit nested
                issuer = dict(x[0] for x in cert['issuer'])
                subject = dict(x[0] for x in cert['subject'])

```

```

        print(f"Issuer    : {issuer.get('organizationName', issuer.get('commonName', 'Not Found'))}")
        print(f"Subject   : {subject.get('commonName', 'Not Found')}")
        print(f"Valid From : {cert.get('notBefore', 'Not Found')}")
        print(f"Valid Until: {cert.get('notAfter', 'Not Found')}")

    summary["SSL"] = True
except Exception as e:
    print(f"[!] Could not retrieve SSL Certificate: {e}")

def get_http_info(domain):
    print_section_header("HTTP Security Headers & Technologies")
    url = f"https://{domain}"
    try:
        response = requests.get(url, timeout=5, allow_redirects=True)
        headers = response.headers

        # Check standard security headers
        sec_headers = [
            'Strict-Transport-Security',
            'Content-Security-Policy',
            'X-Frame-Options',
            'X-Content-Type-Options',
            'Referrer-Policy'
        ]

        for h in sec_headers:
            val = headers.get(h)
            print(f"{h.ljust(25)}: {val if val else 'Not Found'}")

        print("\n[Technologies / Fingerprinting (Best Effort)]")
        server = headers.get('Server')
        x_powered = headers.get('X-Powered-By')
        print(f"Server        : {server if server else 'Unknown'}")
        print(f"X-Powered-By   : {x_powered if x_powered else 'Unknown'}")

        summary["HTTP_Security"] = True
    except requests.RequestException as e:
        print(f"[!] HTTP request failed: {e}")

    print_section_header("Common Web Files")
    try:
        # Check robots.txt
        robots_url = f"{url}/robots.txt"
        r_robots = requests.get(robots_url, timeout=5)
        if r_robots.status_code == 200:
            print("robots.txt    : Exists")
        else:

```

```

        print("robots.txt    : Not Found")

    # Check sitemap.xml
    sitemap_url = f"{url}/sitemap.xml"
    r_sitemap = requests.get(sitemap_url, timeout=5)
    if r_sitemap.status_code == 200:
        print("sitemap.xml    : Exists")
    else:
        print("sitemap.xml    : Not Found")

    summary["Web_Files"] = True
except requests.RequestException:
    print("[!] Could not check common web files due to connection errors.")

def print_summary(execution_time):
    print("\n" + "="*50)
    print("          EXECUTION SUMMARY")
    print("="*50)
    print(f"Total Execution Time: {execution_time:.2f} seconds\n")

    for module, status in summary.items():
        status_text = "SUCCESS" if status else "FAILED"
        print(f"[*] {module.ljust(15)} : {status_text}")
    print("="*50)

def main():
    print(r"""
    _____
   /_/_\_____(_)_____/_/_\_____
  ////_/_\____/___/|_|_V///_/_V___\
 /___//_/_\___/|_|_|_|_|_|_|_|_|_|
/ /_/_\_/___/___/|_|_|_|_|_|_|_|_|_|
Passive Reconnaissance Tool (Educational Use Only)
""")

    domain = input("Enter Domain (e.g., shopify.com): ").strip()

    # Strip protocols if user accidentally pastes a URL
    if domain.startswith("http://") or domain.startswith("https://"):
        domain = urlparse(domain).netloc

    if not domain:
        print("[!] No domain entered. Exiting.")
        return

    print(f"\n[*] Starting Passive Reconnaissance on: {domain}")
    start_time = time.time()

```

```
# Execute Modules
get_whois_info(domain)
get_ip_asn_info(domain)
get_dns_info(domain)
get_ssl_info(domain)
get_http_info(domain)

# End
end_time = time.time()
print_summary(end_time - start_time)

if __name__ == "__main__":
    main()
```

Challenges Faced

- Some tools such as Amass took a long time to run and occasionally returned results that overlapped with Subfinder and Assetfinder, so the results had to be manually merged and de-duplicated.
- Not every discovered subdomain was live, and some live hosts returned redirects or restricted (401/403) responses, which made it harder to confirm the exact technology in use.
- Passive fingerprinting has limits — for some assets the exact programming language and CMS could not be confirmed without more invasive testing, which was outside the scope of this assignment.
- Building the Google Dork list took extra time because many queries returned no useful results and had to be refined before they produced meaningful evidence.
- Keeping the report organized with a large number of screenshots and tables, while staying within the ethical and passive-only scope of the assignment, required careful documentation as the work was carried out.

Conclusion

This assignment gave hands-on experience with the full passive reconnaissance workflow that is used before any authorized penetration test or bug bounty engagement begins.

Across the ten tasks, public information was gathered about the target organizations' domains, DNS records, subdomains, live web assets, and technologies, and the results were organized into a single attack surface picture.

The risk identification task (Task 9) showed that even without any active exploitation, passive reconnaissance alone can reveal a large number of potential security risks, such as exposed admin pages, forgotten subdomains, and missing email security records.

Overall, this exercise reinforced why reconnaissance and OSINT are treated as the first and most important phase of any security assessment, and it improved practical skills in using industry-standard tools such as WHOIS, DNS lookup utilities, Subfinder, Amass, Assetfinder, httpx, Gowitness, and Google Dorking.

References

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- Subfinder (ProjectDiscovery) — <https://github.com/projectdiscovery/subfinder>
- OWASP Amass — <https://github.com/owasp-amass/amass>
- Assetfinder (Tomnomnom) — <https://github.com/tomnomnom/assetfinder>
- httpx (ProjectDiscovery) — <https://github.com/projectdiscovery/httpx>
- Gowitness — <https://github.com/sensepost/gowitness>
- Google Hacking / Dorking reference — Google Advanced Search Operators.
- Wappalyzer / built-in fingerprinting used for technology identification.

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- Python documentation for requests, dnspython, python-whois, and ipwhois libraries.